

CN Semester Examination 2024

Computer Networks (CN) · Semester Examination · 2024

Max. Marks	75	Time	3 hours
------------	----	------	---------

Answer ALL questions from Part A and any FIVE from Part B.

Part A (25 marks)

1. Illustrate OSI Reference Model with a worked example. (5 marks)
2. Discuss the role of Framing in Data Link Layer & MAC. (5 marks)
3. What is meant by IPv4 Addressing? Discuss its advantages and limitations. (5 marks)
4. Discuss the role of UDP in Transport Layer (TCP/UDP). (5 marks)
5. Explain DNS with a neat diagram and a suitable example. (5 marks)

Part B

1. Define OSI Reference Model. Derive or justify the key properties involved. (Unit 1)
2. Explain TCP/IP Model with a neat diagram and a suitable example. (Unit 1)
3. Illustrate Framing with a worked example. (Unit 2)
4. What is meant by Error Detection? Discuss its advantages and limitations. (Unit 2)
5. Write short notes on IPv4 Addressing. (Unit 3)
6. What is meant by Subnetting? Discuss its advantages and limitations. (Unit 3)
7. What is meant by UDP? Discuss its advantages and limitations. (Unit 4)
8. Compare TCP Connection Management with TCP Congestion Control. When would you prefer one over the other? (Unit 4)
9. Discuss the role of DNS in Application Layer & Security. (Unit 5)
10. Write short notes on HTTP. (Unit 5)